Jorge Sánchez Aguilar

AI Engineer | Computer Vision · Agentic Systems · MLOps

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SUMMARY

Final-year CS student and AI Engineer at Ferrovial Digital Hub, where I build production computer vision systems for US highway infrastructure. NASA Space Apps 2025 Global Finalist. Focus: shipping AI systems from model training through production deployment, and multi-agent LLM architectures.

EXPERIENCE

AI Engineer Intern — Digital Hub, Ferrovial — Sept 2025 – Present

Madrid, Spain (Hybrid) · Computer Vision team — US Highway Tolling Projects

- Run the full computer vision workflow end-to-end: dataset curation from raw CCTV footage, model training on Azure ML compute, formal evaluation (F1, precision/recall, confusion matrices), and automated reporting used by the US team to support project decisions.
- Designed and built a night-time vehicle detection model using YOLO26-Pose for low-quality public CCTV with heavy headlight noise; chosen for production over CenterNet and YOLOv8 alternatives developed by the team due to 3x faster inference at comparable accuracy.
- Designed and shipped a non-AI camera perspective-shift detector using geometric heuristics and a 3-check validation gate (2-of-3 required), detecting sub-pixel CCTV movement in real time; selected for production over ML alternatives due to speed and accuracy.
- Stack: Python, PyTorch, OpenCV, Docker, Azure ML, Azure DevOps, Azure Blob Storage, MLflow, Git.

PROJECTS

Multi-Agent LLM System for Financial Market Analysis — March 2026 – Present

- Architected a 3-agent LLM pipeline (MacroAnalyst, TradeAnalyst, RiskReviewer) producing validated trade proposals across 36 assets in 4 asset classes (crypto, equities, ETFs, commodities); each agent is traced end-to-end in Langfuse for full observability of prompts, tokens, and latency.
- Deployed local inference with Qwen3-32B-AWQ on vLLM via Docker Compose with health-checked services and hardware profiles; provider-agnostic routing (LLM_BASE_URL override) allows switching between local vLLM and OpenRouter without code changes.
- Engineered a typed data layer with Pydantic v2 frozen contracts at every module boundary, SQLite with versioned migrations and full signal lifecycle tracking, and a fully-mocked unit test suite running in GitHub Actions CI with pytest, ruff, and mypy on every push.

Space Atlas — NASA Space Apps Challenge — Oct 2025

Team of 5 · Madrid Winner · Global Finalist

- Built the Node.js/Express backend and deployed it to Azure App Service with a GitHub Actions CI/CD pipeline, integrating Azure OpenAI (GPT-4o) with Azure Blob Storage through a REST API that generates real-time educational explanations of Martian geological features from high-resolution imagery.

EDUCATION

B.S. Computer Science — Universidad Rey Juan Carlos — Expected July 2026

Concentrations: Artificial Intelligence, Modeling & Simulation. Madrid, Spain.

42 Madrid — Peer-to-peer software engineering program — 2023 – 2025

Level 3 (80% complete). Focus: C/C++ systems programming, UNIX/Linux, object-oriented design, AI/ML.

AWARDS & COMPETITIONS

- **NASA Space Apps Challenge 2025 — Madrid Winner · Global Finalist**. Advanced to global finalist round out of 11,500+ teams and 114,000+ participants worldwide.
- **Qubic × Vottun Madrid Hackathon 2025 — Top 10 Finalist**

SKILLS

AI / ML: Computer Vision, Object Detection, Pose Estimation, LLMs, Multi-Agent Systems, Prompt Engineering, vLLM, Hugging Face Transformers, NLP.

Tools & Cloud: Docker, Git, GitHub Actions (CI/CD), REST APIs, Azure (OpenAI, ML, Blob Storage, App Service, DevOps), Langfuse, MLflow, Linux/UNIX.

Languages: English (C2 Proficiency), Spanish (Native).